PART 2

1

EXPERIMENT 1

Study of Mitosis and Meiosis

Aim

To study different stages of mitosis and meiosis.

I. STUDY OF MITOSIS

Requirements

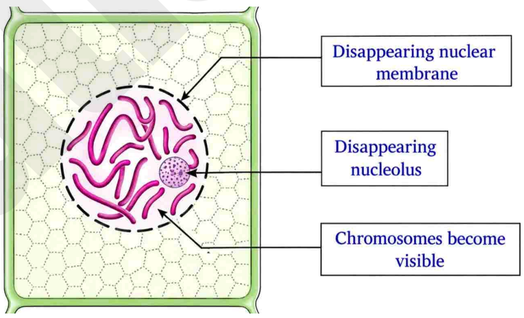
Permanent slides of onion root tips showing mitosis, and a compound microscope.

Procedure

Focus the permanent slide of the squash preparation of an onion root tip under the high power of the microscope, and observe the changes taking place in the nucleus through the following stages of mitosis.

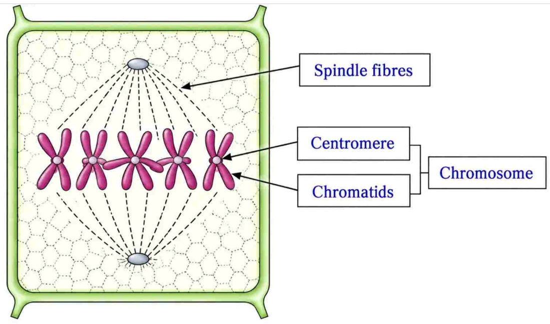
1. Prophase

1. In prophase chromosomes are seen in the form of long and coiled thread-like structures.
2. Each chromosome consists of two sister chromatids connected by centromere.
3. The nuclear membrane and nucleolus seen distinctly during the early prophase are seen disintegrating and disappearing towards the end of the prophase.



2. Metaphase

1. Nuclear membrane disappears completely in metaphase. The chromosomes are condensed and shortened and therefore they are seen more distinct and clear.
2. The formation of spindle fibres is completed during metaphase.



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(3) Where are spindle fibres formed?

Spindle fibres are formed between centromere of each chromosome and both centrioles of opposite poles.

(4) How are chromosomes arranged during this phase?

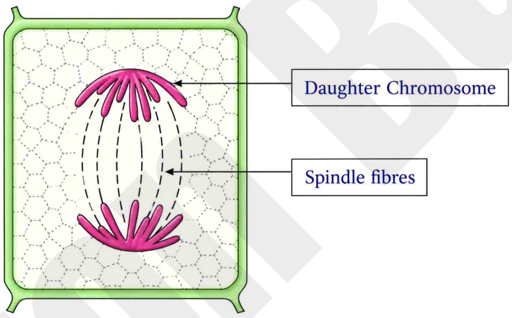
The chromosomes are seen arranged at equatorial plane of the spindle, during this phase.

3. Anaphase

(1) Which important division process takes place during anaphase?

In this phase, centromere split and sister chromatids of each chromosome separate and move towards the opposite poles due to the contraction of spindle fibres.

2. The chromosomes appear 'L' or 'V' shaped during anaphase.



4. Telophase

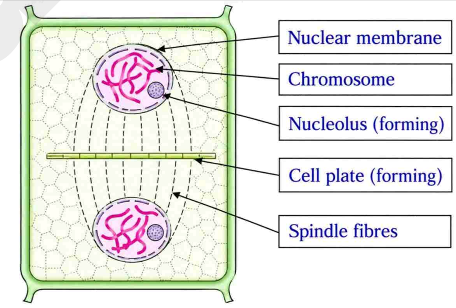
(1) What change takes place in the chromosomes during this phase?

The chromosomes are seen assembled at the poles in the form of long coiled thread-like structures.

(2) What are important changes during telophase?

During telophase the chromosomes reorganise to form two daughter nuclei at opposite poles.

3. The nuclear membrane and nucleolus reappear at the end of telophase forming two daughter nuclei. The process of cytokinesis takes place at the end forming two daughter cells.



II. STUDY OF MEIOSIS

Meiosis is also called reduction division because the chromosome number is reduced to half.

Requirements

Slides of grasshopper testis showing meiosis, and a compound microscope.

Procedure

Focus the permanent slides of meiosis under the high power of a compound microscope. Observe the slides carefully and note down the details.

Observations

(1) What are the main divisions of the meiosis?

(A) Meiosis I (B) Meiosis II

(2) Which is the most elaborate phase of the meiosis I? What are its substages?

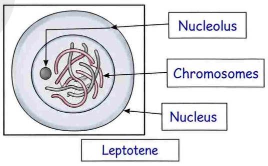
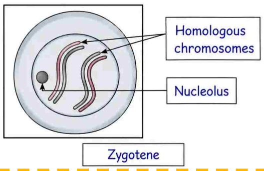
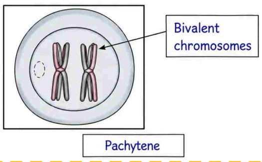
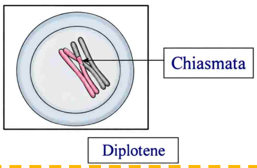
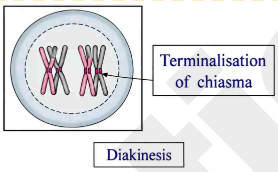
Prophase 'I' is the most elaborate phase of the meiosis I.

5 substages of prophase I are as follows: Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.

3. At the end of Meiosis I, diads, i.e. 2 distinct cells are visible. The chromosome number is reduced to half.
4. At the end of Meiosis II, tetrads, i.e. 4 distinct daughter cells are visible.

Meiosis I

Identify the diagrams of different phases of Prophase-I and name them. Label the different parts of each phase and write at least two characteristics of each phase.

Prophase – I	Characteristics
Leptotene 	Chromosomes appear as thin and long threads. Nuclear envelope and nucleolus are distinct.
Zygotene 	Homologous chromosomes form pairs. They are known as bivalents. This pairing is called synapsis.
Pachytene 	The chromosomes clearly show two similar chromatids. A bivalent becomes tetrad. Exchange of chromatid segments between homologous chromosomes takes place. This is known as crossing over. Recombination nodule or chiasmata are formed.
Diplotene 	The recombined homologous chromosomes start separating. They remain attached only at the points of crossing over i.e., chiasmata.
Diakinesis 	Terminalisation of chiasmata is complete. Chromosomes appear as thick 'X', 'L' or ring shaped ('o') structures. Nucleolus and nuclear envelope disappears.

Write the important events occurring in each of the remaining phases after Prophase I:

Metaphase I	Anaphase I
Bipolar spindle is formed. Homologous chromosomes (bivalents) arranged at equatorial plane by centromere.	The homologous chromosomes are pulled away from each other. They move to the opposite poles.
Telophase I	**Cytokinesis I**
Chromosomes reach to the respective poles. Nucleolus and nuclear envelope reappear. The cell shows two daughter nuclei.	A cell plate or furrow appears in the middle and extends on its side. Two haploid daughter cells are formed.

Meiosis II

Write the important events occurring in the phases of Meiosis II.

Prophase II	The chromosomes of daughter cells become thick and distinct. Nucleolus and nuclear envelope disappear.
Metaphase II	Bipolar spindle formation takes place in both the cells. The chromosomes get attached at the equator of the spindle.
Anaphase II	The sister chromatids separate. The separated chromatids (daughter chromosomes) start migrating towards the opposite poles. The separation occurs at the centromere.
Telophase II	Daughter chromosomes reach poles. The nucleolus and nuclear envelope reappear. Two daughter nuclei are now clearly seen in each cell. These will later form 4 cells. Thus, total 4 cells are formed from original one cell.
Cytokinesis II	Formation of cell plate takes place in plant cells. Animal cell divides by formation of a constriction in plasma membrane. At the end, four daughter cells are formed, each with haploid number of chromosomes.

Inferences

1. By mitotic division, cells multiply in number without changing the chromosome number.
2. Meiosis is reduction division in which chromosome number of the cell is reduced to half.

Multiple Choice Questions

1. Which type of cells show mitotic cell division?

(A) Somatic (B) Germinal
(C) Somatic and Germinal (D) None of the above

Answer: (C) Somatic and Germinal

2. In which phase of mitosis, the chromosomes are at the equatorial plane?

(A) Prophase (B) Metaphase (C) Anaphase (D) Telophase

Answer: (B) Metaphase

3. How many cells can be formed if cell divides 6 times by mitosis?

(A) 32 (B) 44 (C) 64 (D) 72

Answer: (C) 64

4. Which of the following material can be used to observe meiotically dividing cells?

(A) Onion root tip (B) Grasshopper testis
(C) Potato starch grains (D) Cheek cells of human

Answer: (B) Grasshopper testis

5. Which one is the most important difference in mitosis and meiosis?

(A) Number of daughter cells formed (B) Number of chromosomes in daughter cells
(C) Amount of cytoplasm in daughter cells (D) Time required for the cell division

Answer: (B) Number of chromosomes in daughter cells

Remember

1. Mitosis take place both in somatic cells and germinal cells.
2. Meiosis takes place only in germinal cell. It does not take place in somatic cells.

Date: ______________ Teacher's Signature: ______________